

MONAI for Clinical DICOM Workflows

Hitendrasinh Rathod

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1 Overview

MONAI is widely adopted for medical imaging research and model development, often using curated datasets stored in formats such as NIfTI. In real-world healthcare environments, however, CT and MRI data are generated, stored, and exchanged primarily as DICOM and integrated into PACS and RIS workflows.

This document provides a clinical-ready reference example for working directly with DICOM CT and MRI data in MONAI, with considerations relevant to hospital and NHS-style radiology environments. The aim is to help practitioners bridge the gap between research-oriented pipelines and deployable, production-aware AI systems, while maintaining safety, reproducibility, and alignment with clinical practice.

This example focuses exclusively on preprocessing and intentionally excludes model training, benchmarking, or patient-identifiable data.

2 Intended Audience

This document is intended for:

- Healthcare AI engineers transitioning MONAI pipelines from research to production
- Clinical data scientists working with PACS-integrated imaging data
- NHS and hospital teams building AI decision-support systems for radiology

It is not intended as a training tutorial or a certified implementation of medical devices, but rather as a reference preprocessing template aligned with real clinical workflows.

3 Clinical Workflow Context

In production radiology environments, AI systems typically operate within workflows similar to the following steps:

1. Imaging data is acquired via CT or MRI scanners and stored as DICOM series in PACS.
2. The DICOM series is retrieved from PACS for AI processing.
3. Pre-processing steps, such as intensity scaling, normalization, and channel formatting, are applied.
4. AI models generate outputs (e.g., segmentation, detection, or classification maps).
5. Outputs are returned to the radiology workflow as decision support, integrated into PACS/RIS.
6. Auditing and logging ensure reproducibility and compliance with clinical governance.

Typical characteristics of these environments include:

- Imaging data stored and exchanged as DICOM series

- CT images encoded in Hounsfield Units (HU) requiring clinically meaningful windowing
- MRI images with non-standardized intensity scales
- Strict requirements for information governance, auditability, and anonymisation
- AI outputs used as decision support rather than autonomous diagnosis

4 DICOM Loading in MONAI

MONAI supports DICOM ingestion via its image readers, leveraging ITK and pydicom internally. In clinical workflows, DICOM series are commonly loaded at runtime rather than pre-converted to NIfTI, enabling closer alignment with PACS-based pipelines and reducing preprocessing assumptions.

5 Example: CT DICOM Preprocessing with Clinical Windowing

CT imaging uses a standardized intensity scale (Hounsfield Units), allowing clinically grounded windowing for different diagnostic tasks such as lung and soft tissue evaluation.

```
from monai.transforms import (
    LoadImage,
    EnsureChannelFirst,
    ScaleIntensityRange,
    Compose
)

# Example preprocessing pipeline for CT DICOM data
ct_transforms = Compose([
    LoadImage(image_only=True),
    EnsureChannelFirst(),
    ScaleIntensityRange(
        a_min=-1000,
        a_max=400,
        b_min=0.0,
        b_max=1.0,
        clip=True
    )
])

# Example usage
ct_image = ct_transforms("/path/to/dicom/series")
```

Listing 1: CT preprocessing pipeline

Notes:

- Direct ingestion of DICOM series without format conversion
- Channel-first formatting required by MONAI networks
- Clinically meaningful CT intensity normalization
- Suitable for inference-time pipelines

6 MRI Preprocessing Considerations

MRI intensity values vary across scanners and protocols. Recommended principles:

- Avoid assumptions based on Hounsfield Units
- Apply sequence-aware preprocessing (e.g., T1, T2, STIR)
- Prefer percentile-based or z-score normalization

```
from monai.transforms import (
    LoadImage,
    EnsureChannelFirst,
    NormalizeIntensity,
    Compose
)

mri_transforms = Compose([
    LoadImage(image_only=True),
    EnsureChannelFirst(),
    NormalizeIntensity(nonzero=True)
])

mri_image = mri_transforms("/path/to/dicom/series")
```

Listing 2: MRI preprocessing pipeline

7 Deployment Notes

- Anonymize patient metadata before processing
- Ensure reproducibility and auditability
- Validate across vendors and protocols
- Treat AI outputs as decision support

8 Why This Example Matters

Real hospital environments use DICOM directly from PACS. This example bridges the gap between research pipelines and production-ready clinical AI workflows. It also accompanies a **code module** (`monai.transforms.clinical_preprocessing.py`) and an **example notebook** (`tutorials/clinical_dicom_pipeline.ipynb`) for practical usage.

9 References

- MONAI Documentation: <https://monai.io>
- DICOM Standard: <https://www.dicomstandard.org>